Evaluating the Interpretability and Usefulness of LIME and SHAP Explanations for Architecture Technical Debt (ATD) Classification from Jira Issues

Transformer-based models are inherently opaque, making their predictions difficult to interpret. LIME and SHAP offer post-hoc explanations to improve model transparency.

The goal of this survey is to gather expert feedback on the interpretability and usefulness of explanations produced by Transformer models for ATD detection.

Instructions:

1. You will be shown several explanation outputs generated by LIME and SHAP for actual ATD classification cases
2. Please provide your individual assessment on each output
3. All responses are anonymous, and results will be summarized

Your responses are confidential and will only be accessed by authorized researchers. We ensure strict data protection measures to maintain the anonymity and security of your information. The survey will take approximately 10-15 minutes. Thank you for participating in this survey and supporting our research.

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* Indicates required question

Section A: Demographics and Expertise

1. Please indicate your primary area(s) of expertise (check all that apply): *

Check all that apply.

- [ ] Technical Debt
- [ ] Self-Admitted Technical Debt
- [ ] Software Architecture
- [ ] Other: _____

2. How familiar are you with Local Interpretable Model-agnostic Explanations (LIME) XAI methods? *

Mark only one oval.

- ( ) Not at all
- ( ) Somewhat
- ( ) Very familiar

3. How familiar are you with SHapley Additive exPlanations (SHAP) XAI methods? *

Mark only one oval.

- ( ) Not at all
- ( ) Somewhat
- ( ) Very familiar

Section B: Model Output Without Explanation

The following is a model prediction for a Jira issue, **without** any explanation from XAI methods. Please review the output and answer the questions below.

This example is based on the model's classification of Jira issue **CAMEL-19998**.

The issue (summary + description) below is classified as **ATD**

Camel / CAMEL-19998

camel-core: cleanup cyclic dependencies in the AbstractCamelContext

Details

Description

As part of ~~CAMEL-15105~~ we decoupled the ExtendedCamelContext from the AbstractCamelContext. This helped us reduce the incidence of type polution problems, cleanup and simplify the code.

However, there's still many cases of cyclic dependencies between these two types. We should move the handling of the internal APIs fully to the extension class.

4. **1. Trust in the Model's Output:** *

To what extent does seeing only the predicted class increase your trust in the model's classification decision?

Mark only one oval.

- Not at all
- Slightly
- Moderately
- Very much
- Completely

5. 1. (Optional) Please elaborate on your answer:

6. **2. Causality:** *

Without any explanation or highlighted features, how well can you identify which factors contributed to the model's decision to classify this issue as ATD?

Mark only one oval.

- Very unclear
- Somewhat unclear
- Neutral
- Clear
- Very clear

7. 2. (Optional) Please elaborate on your answer:

8. **3. Transferability** *

To what extent does the information provided (**predicted class**) help you apply the model's reasoning to similar issues in other projects or contexts?

Mark only one oval.

- Not at all
- Slightly
- Moderately
- Very much
- Completely

9. 3. (Optional) Please elaborate on your answer:

10. **4. Informativeness** *

Does the output (predicted class) provide enough information for you to understand why the model classified this issue as ATD?

Mark only one oval.

- Strongly disagree
- Disagree
- Neutral
- Agree
- Strongly agree

11. 4. (Optional) What, if anything, was missing or misrepresented?

12. **5. Confidence** *

To what extent does the predicted class alone give you confidence in the model's decision?

Mark only one oval.

- Not at all
- Slightly
- Moderately
- Very much
- Completely

13. 5. (Optional) Please elaborate on your answer:

14. **6. Fairness** *

Does seeing only the predicted class allow you to evaluate whether the model's classification might be biased or unfair?

Mark only one oval.

- Not at all
- Slightly
- Moderately
- Very much
- Completely

15. 6. (Optional) Please elaborate on your answer:

16. **7. Accessibility** *

Is the model output (predicted class) understandable and useful to a software architect or developer with no background in machine learning?

Mark only one oval.

- Strongly disagree
- Disagree
- Neutral
- Agree
- Strongly agree

17. 7. (Optional) How could this explanation be made more actionable?

18. **8. Interactivity** *

Would your understanding of the model's decision be improved if you could interact with the output (for example, by seeing which factors influenced the decision or by adjusting input values)?

Mark only one oval.

- Not at all
- Slightly
- Moderately
- Very much
- Completely

19. 8. (Optional) Please elaborate on your answer:

Section B: LIME Evaluation of Explanations

The picture below shows a **LIME** explanation of a prediction generated by the BERT language model that was applied to the problem of ATD detection from Jira issues. The visualization tries to present the factors that most significantly influenced the prediction. The following short questions ask you to identify the elements of the visualizations you find the most informative.

The figure below is produced from the Apache Camel Jira issue **CAMEL-19998.**

LIME XAI (You can right-click the figure and select "Open image in new tab" to see a clearer version)

Text with highlighted words

camel-core: cleanup cyclic dependencies in the AbstractCamelContext - As part of CAMEL-15105 we decoupled the ExtendedCamelContext from the AbstractCamelContext. This helped us reduce the incidence of type polution problems, cleanup and simplify the code.

However, there's still many cases of cyclic dependencies between these two types. We should move the handling of the internal APIs fully to the extension class.

20. **1. Trust in the Model's Output:** *

To what extent does the combination of prediction probability scores (Non-ATD: 0.17 and ATD: 0.83), highlighted the most influential tokens and **LIME** bar charts, increase your trust in the model's decision to classify this issue as ATD?

Mark only one oval.

- Not at all
- Slightly
- Moderately
- Very much
- Completely

21. 1. (Optional) Please elaborate on your answer:

22. **2. Causality:** *

Do the bar chart and highlighted tokens in the explanation help you identify which specific features (words) most influenced the model's decision for ATD classification?

Mark only one oval.

- Very unclear
- Somewhat unclear
- Neutral
- Clear
- Very clear

23. 2. (Optional) Please elaborate on your answer:

24. **3. Transferability** *

Based on the information provided (prediction probabilities, influential tokens, and highlighted text), to what extent do you feel you could apply the model's reasoning to similar issues in other projects or contexts?

Mark only one oval.

- Not at all
- Slightly
- Moderately
- Very much
- Completely

25. 3. (Optional) Please elaborate on your answer:

26. **4. Informativeness of Highlighted Tokens:** *

Do the highlighted tokens (e.g., "**dependency**", "**decouple**", "**move**", etc) correspond to what you would personally consider indicators of architectural technical debt in this context?

Mark only one oval.

- Strongly disagree
- Disagree
- Neutral
- Agree
- Strongly agree

27. 4. (Optional) What, if anything, was missing or misrepresented?

28. **5. Prediction Probability Score (Confidence)** *

Does the probability score visualization (Non-ATD: 0.17, ATD: 0.83) provide you with a meaningful sense of the model's confidence in its classification, beyond simply showing the final label?

Mark only one oval.

- Not at all
- Slightly
- Moderately
- Very much
- Completely

29. 5. (Optional) Please elaborate on your answer:

30. **6. Fairness** *

Does the explanation help you detect potential biases or unfair influences in the model's classification decision (e.g., overemphasis on certain terms, issue types, or components)?

Mark only one oval.

- Not at all
- Slightly
- Moderately
- Very much
- Completely

31. 6. (Optional) Please elaborate on your answer:

32. **7. Accessibility of Explanation** *

Is the explanation—through the use of bar charts, prediction probabilities, and highlighted text—clear and understandable to someone who may not have expertise in machine learning?

Mark only one oval.

- Strongly disagree
- Disagree
- Neutral
- Agree
- Strongly agree

33. 7. (Optional) How could this explanation be made more actionable?

34. **8. Interactivity** *

To what extent would the ability to interact with this explanation (e.g., changing feature values or exploring alternative scenarios) improve your understanding of the model's decision?

Mark only one oval.

- Not at all
- Slightly
- Moderately
- Very much
- Completely

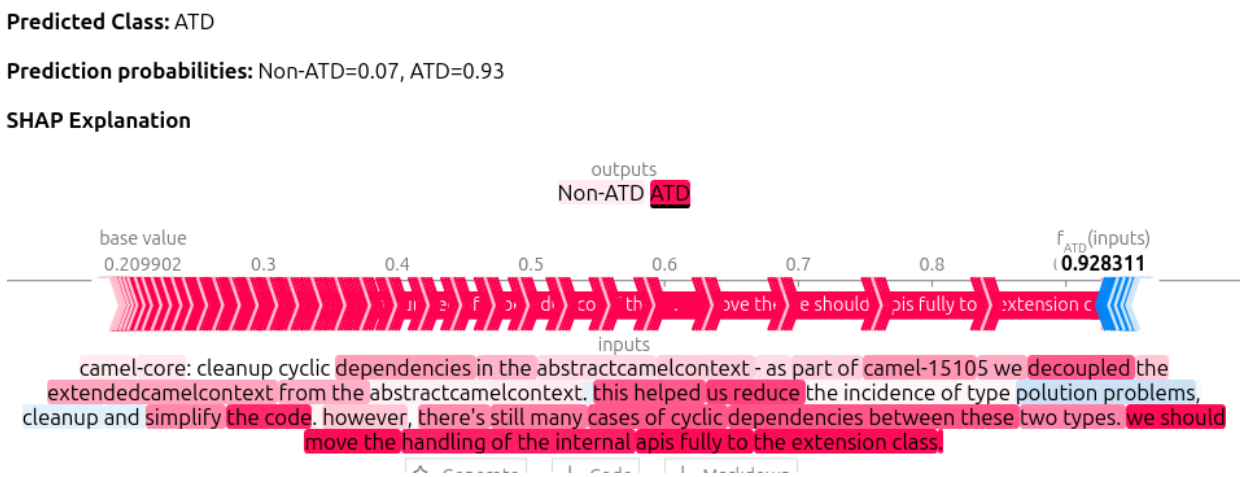
35. 8. (Optional) Please elaborate on your answer:

Section B: SHAP Evaluation of Explanations

The picture below shows a **SHAP** explanation of a prediction generated by the BERT language model that was applied to the problem of ATD detection from Jira issues. The visualization tries to present the factors that most significantly influenced the prediction. The following short questions ask you to identify the elements of the visualizations you find the most informative.

The figure below is produced from the Apache Camel Jira issue **CAMEL-19998**.

SHAP XAI (You can right-click the figure and select "Open image in new tab" to see a clearer version)



36. **1. Trust in the Model's Output:** *

To what extent does the combination of the predicted class, probability scores (Non-ATD: 0.07 and ATD: 0.93), the highlighted text segments, and **SHAP** plot increase your trust in the model's decision to classify this issue as ATD?

Mark only one oval.

- Not at all
- Slightly
- Moderately
- Very much
- Completely

37. 1. (Optional) Please elaborate on your answer:

38. **2. Causality:** *

Do the arrows and highlighted text in the **SHAP** plot help you identify which specific features (words or phrases) most influenced the model's decision for ATD classification?

Mark only one oval.

- Very unclear
- Somewhat unclear
- Neutral
- Clear
- Very clear

39. 2. (Optional) Please elaborate on your answer:

40. **3. Transferability** *

Given the visual display of feature impacts and highlighted text, to what extent do you feel you could apply the model's reasoning to similar issues in other projects or contexts?

Mark only one oval.

- Not at all
- Slightly
- Moderately
- Very much
- Completely

41. 3. (Optional) Please elaborate on your answer:

42. **4. Informativeness of Highlighted Tokens:** *

Do the highlighted tokens (e.g., "**dependencies**", "**move**", "**simplify the code**", "**extension class**") correspond to what you would personally consider indicators of ATD in this context?

Mark only one oval.

- Strongly disagree
- Disagree
- Neutral
- Agree
- Strongly agree

43. 4. (Optional) What, if anything, was missing or misrepresented?

44. **5. Prediction Probability Score (Confidence)** *

Does the probability score shown by SHAP (Non-ATD: 0.07, ATD: 0.93), in combination with the visual explanation, give you a meaningful sense of the model's confidence in its classification, beyond simply showing the final label??

Mark only one oval.

- Not at all
- Slightly
- Moderately
- Very much
- Completely

45. 5. (Optional) Please elaborate on your answer:

46. **6. Fairness** *

Does the explanation help you detect potential biases or unfair influences in the model's classification decision (e.g., overemphasis on certain terms, issue types, or components)?

Mark only one oval.

- Not at all
- Slightly
- Moderately
- Very much
- Completely

47. 6. (Optional) Please elaborate on your answer:

48. **7. Accessibility of Explanation** *

Is the explanation—through the use of color-coded arrows, prediction probabilities, and highlighted text—clear and understandable to someone who may not have expertise in machine learning?

Mark only one oval.

- Strongly disagree
- Disagree
- Neutral
- Agree
- Strongly agree

49. 7. (Optional) How could this explanation be made more actionable?

50. **8. Interactivity** *

To what extent would the ability to interact with this explanation (e.g., changing feature values or exploring alternative scenarios) improve your understanding of the model's decision?

Mark only one oval.

- Not at all
- Slightly
- Moderately
- Very much
- Completely

51. 8. (Optional) Please elaborate on your answer:

Section C: Open Feedback

52. **Explainability Preference:** *

Which method do you find more interpretable for this task?

Mark only one oval.

- LIME
- SHAP
- Both equally
- Neither

53. **Missing Elements:**

What information, if any, would help you better understand or trust the model's explanation?

54. **Suggestions for Improvement:**

How can the current explanation method (e.g., token importance, visualization) be improved for evaluating ATD?

55. **Additional Comments:**
Any additional feedback on the use of XAI in ATD detection in Jira issues?

Thank you for your thoughtful and valuable feedback. Your input will help guide future research and the development of tools for explainable ATD detection.

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